12.2 Multilayer Neural Networks

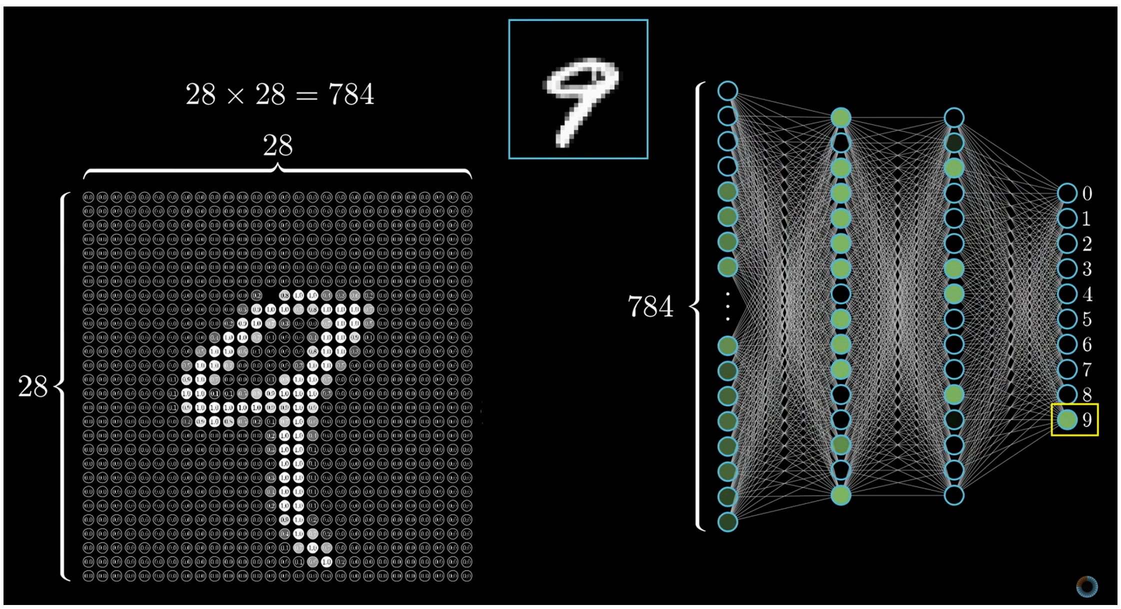
Hidden layers and a composition of linear maps and nonlinearities.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

Figures and notes follow Géron, *Hands-On Machine Learning*; Deisenroth, Faisal, and Ong, *Mathematics for Machine Learning*; Theodoridis; and 3blue1brown.

1. A multi-layer net

A **multi-layer perceptron** stacks dense layers. Below: pixels of a digit flatten to an input vector, then hidden layers, then class scores.



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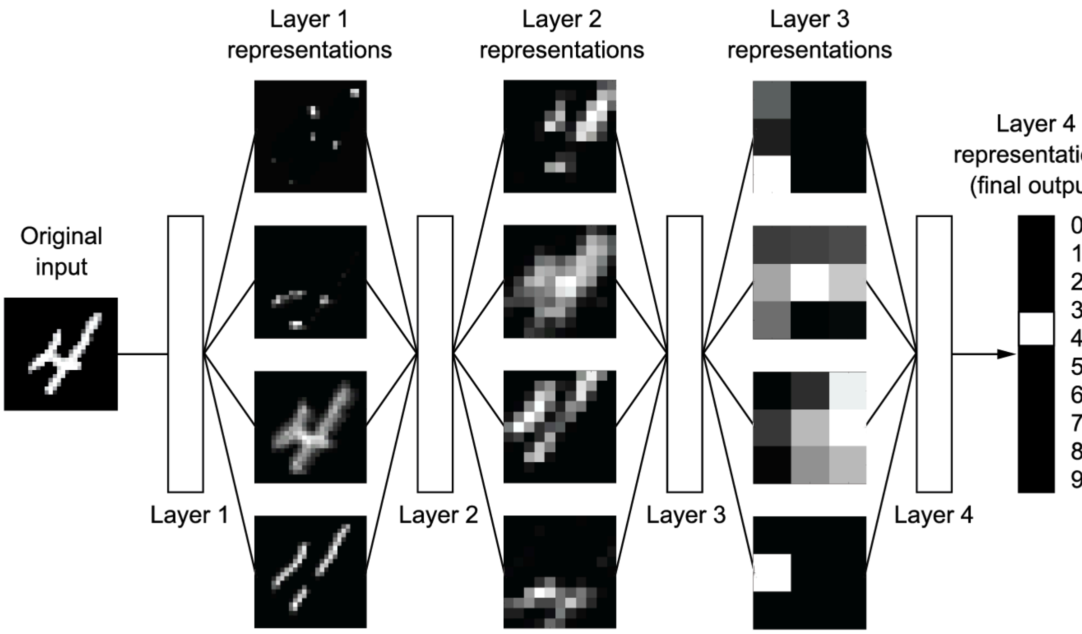
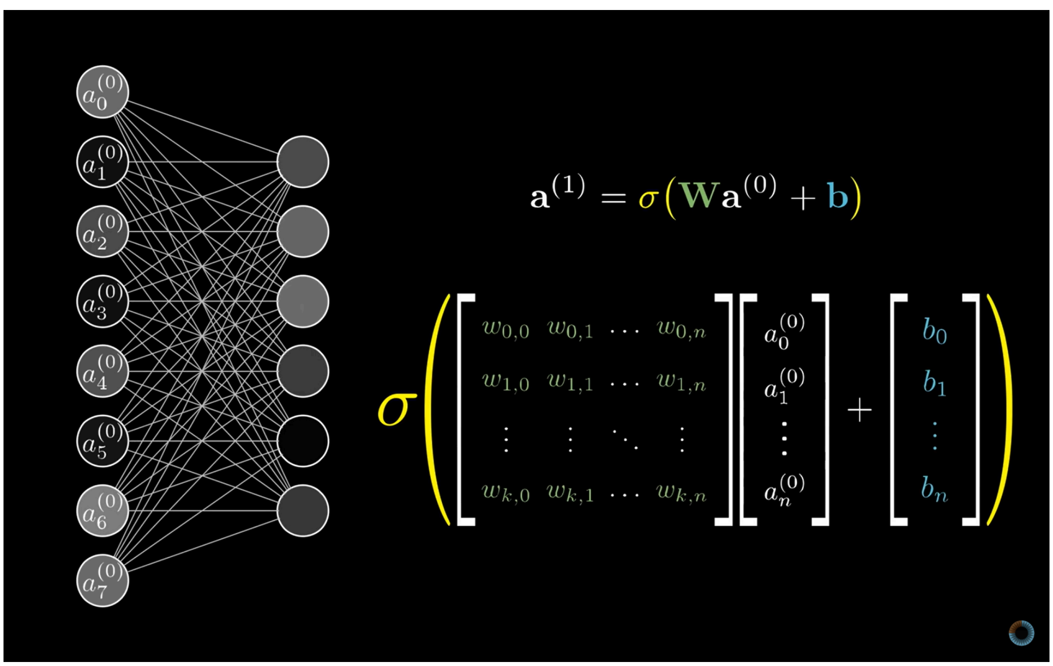
2. What the layers are doing

Think of a deep net as **multistage distillation**: each layer filters the previous representation; what remains is more useful for the task.



3. Linear-algebra notation

Weights, activations, and the next layer line up as matrix–vector products. The picture is the notation you will write.



4. Mathematics of a layer

Linear map. Neuron l does not see raw x after the first layer. It sees the previous activations $a^{(l-1)}$:

$z^{(l)} = W^{(l)}a^{(l-1)} + b^{(l)}$,

where $z^{(l)}$ is the pre-activation, $W^{(l)}$ the weight matrix, $b^{(l)}$ the bias.

Activation. Elementwise nonlinearity:

$$a^{(l)} = \sigma\left(z^{(l)}\right).$$

Without σ, stacked layers collapse to one linear map. The nonlinearity is what a hidden layer is for.

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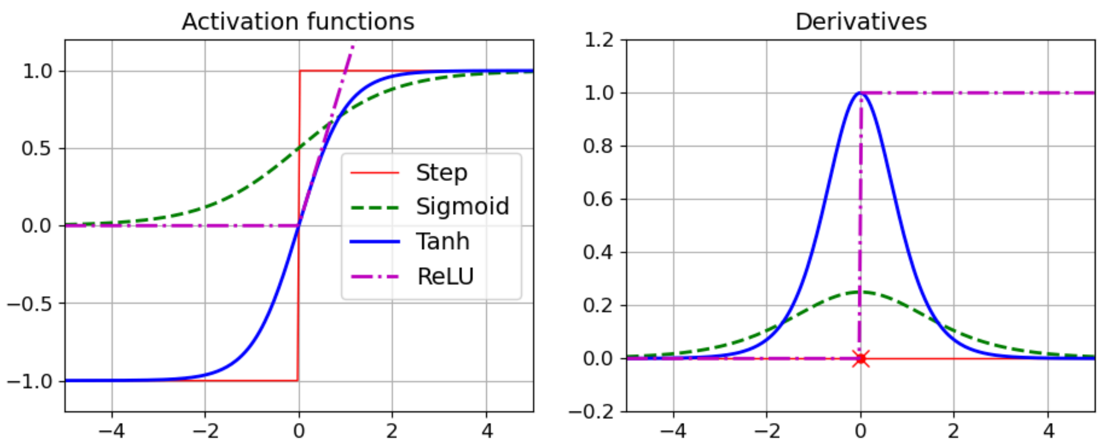
5. Activation functions

Common choices: **sigmoid**, **tanh**, **ReLU**, **softmax**. Applied elementwise after the linear map (softmax on the output for mutually exclusive classes).



Practice

1. Write the forward pass of a two-layer net in one line of matrix products.
2. If you drop σ from every hidden layer, what class of functions can the net represent?
3. In $z^{(l)} = W^{(l)}a^{(l-1)} + b^{(l)}$, what is $a^{(0)}$?